

Total Quality Management — Introduction

Total Quality Management (TQM) is a management approach that aims for long-term success through customer satisfaction. TQM is based on the participation of all members of an organization in improving processes, products, services, and the culture in which they work.

Key idea: quality is everyone's responsibility, not just the quality department's.

TQM originated from the work of W. Edwards Deming and Joseph Juran in post-war Japan and was later adopted worldwide.

Principles of TQM

The main principles of Total Quality Management are:

1. Customer Focus — understanding and meeting customer needs is the primary goal.
2. Total Employee Involvement — every employee participates in working toward common goals.
3. Process-Centered Thinking — a fundamental part of TQM is a focus on process thinking and continuous improvement.
4. Integrated System — all departments and functions must be interconnected with horizontal processes.
5. Strategic and Systematic Approach — a strategic plan must integrate quality as a core component.
6. Continual Improvement — a major thrust of TQM is continuous process improvement, often using PDCA (Plan-Do-Check-Act).
7. Fact-Based Decision Making — decisions are based on data analysis, not opinions.
8. Communication — effective communication ensures morale and motivation at all levels.

TQM Tools and Techniques

Common tools used in TQM include:

- Pareto Charts: identify the most significant factors in a data set.
- Cause-and-Effect (Fishbone) Diagrams: visualize potential causes of a problem organized by category.
- Control Charts: monitor process stability over time.
- Histograms: display frequency distribution of data.
- Scatter Diagrams: show correlation between variables.
- Flowcharts: map out a process step by step.
- Check Sheets: structured forms for collecting and analyzing data.

These are sometimes called the "Seven Basic Tools of Quality".